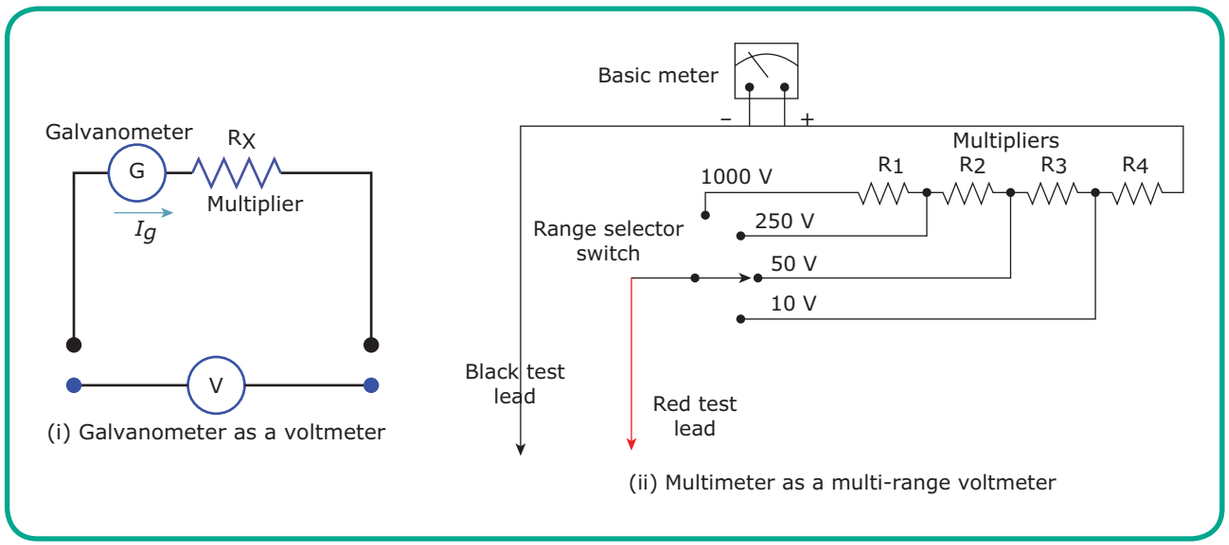


Figure 10.6 Multimeter as a Voltmeter - circuit

10.2.2.2. Multimeter as a Voltmeter

A meter used for measuring voltage is called as voltmeter.

To measure accurate voltage values, various voltage ranges are incorporated in the meter. To measure multiple ranges of voltage, resistors (R_1, R_2, R_3, & R_4) are connected in series (called as multipliers) with the basic meter as shown in the Figure 10.6. With the help of selector switch, we can select a particular range of interest. After selecting the range, the voltage in a circuit can be measured. To measure various ranges of AC and DC voltage measurements, separate multipliers are used in series with the meter.

Note: In general, a galvanometer (basic meter) does not work while measuring AC voltage or current. It works in DC source only.

If AC source is applied to a galvano meter, the needle of the meter tends to oscillate either in the clockwise or anti-clockwise direction with respect to zero position. As a result, there is no movement of the needle takes place and the needle rests near zero position. So, a rectifier is connected in series with the meter to measure AC voltage and current, separately.

10.2.2.3. Multimeter as an Ammeter

An instrument used for measuring current is called Ampere meter shortly Ammeter.

When low resistance is connected in parallel with a galvanometer, it becomes an Ammeter.

Figure 10.7 shows a low value resistance R_S (generally called shunt resistance) connected in parallel with the galvanometer coil. Since the resistance of the coil is higher than the shunt resistance, the majority current is passing through the coil of the meter. The shunt resistance R_S protects the coil from over-current.

In practice, shunt resistances (such as R_1, R_2, R_3, & R_4) are connected in parallel with the meter through selector switch

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